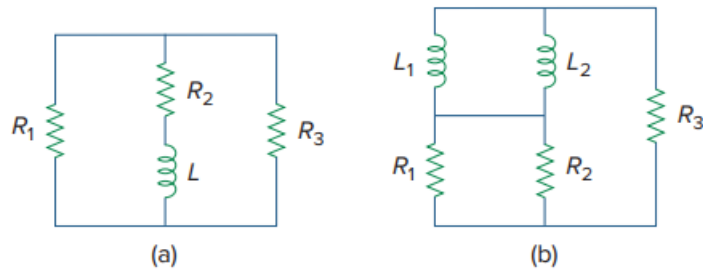


## Problem

Determine the time constant for each of the circuits in Fig. 7.96.

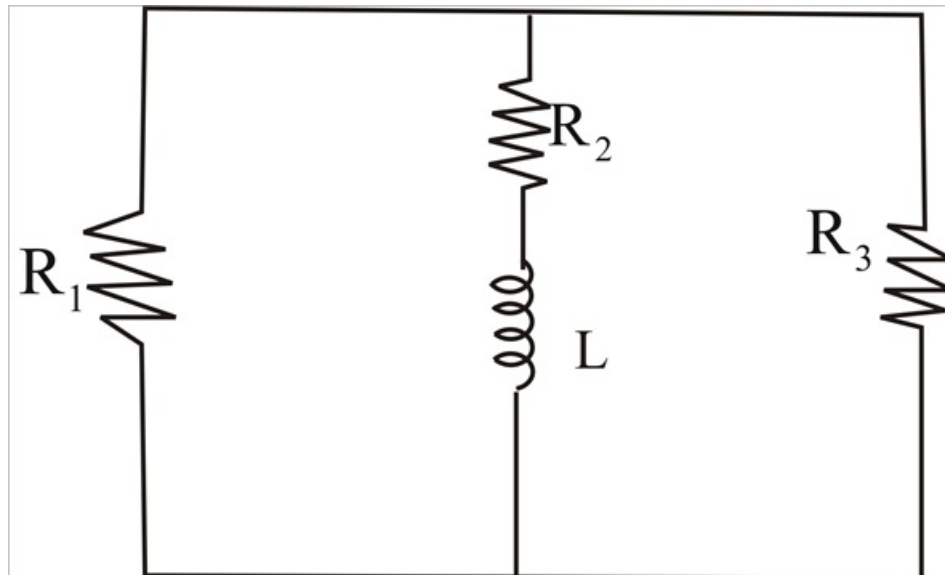
Fig. 7.96:



## Step-by-step solution

## Step 1 of 4

(A)



## Step 2 of 4

To find the equivalent resistance, open the inductor and calculate the resistance through it.

$$R_{eq} = R_2 + (R_1 \parallel R_3)$$

$$R_{eq} = R_2 + \frac{R_1 R_3}{R_1 + R_3}$$

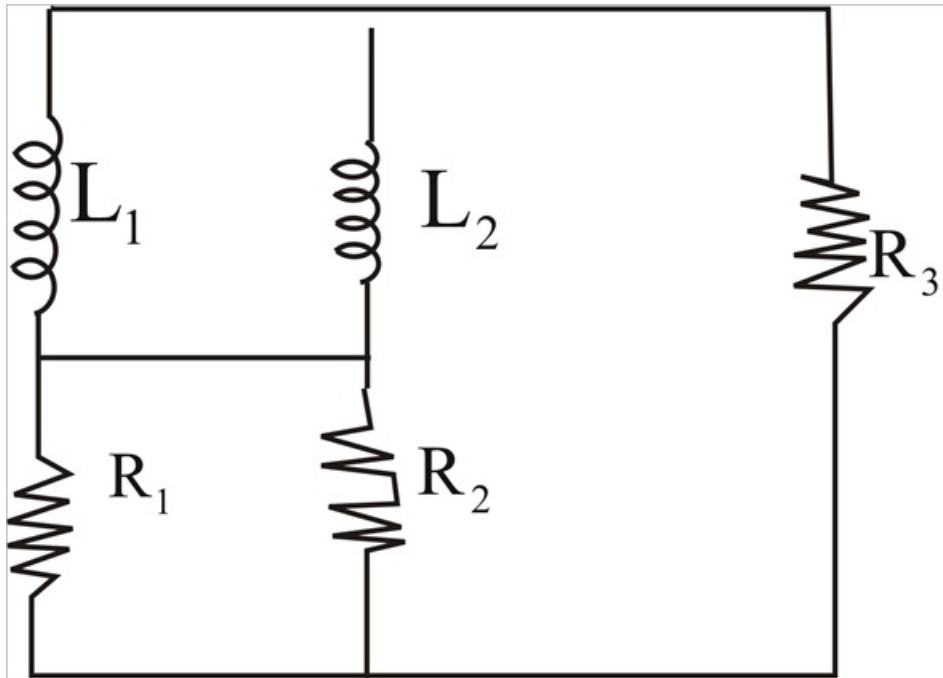
$$R_{eq} = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1 + R_3}$$

So

$$\tau = \frac{L}{R_{eq}}$$

$$\tau = \frac{L(R_1 + R_3)}{R_1 R_2 + R_2 R_3 + R_1 R_3}$$

(B)



## Step 4 of 4

$$L_{eq} = (L_1 \parallel L_2)$$

$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

To find the equivalent resistance, open the inductor and calculate the resistance through it.

$$R_{eq} = R_3 + (R_1 \parallel R_2)$$

$$R_{eq} = R_3 + \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{eq} = \frac{R_1 R_3 + R_2 R_3 + R_1 R_2}{R_1 + R_2}$$

So

$$\tau = \frac{L_{eq}}{R_{eq}}$$

$$\tau = \frac{\left( \frac{L_1 L_2}{L_1 + L_2} \right)}{\left( \frac{R_1 R_3 + R_2 R_3 + R_1 R_2}{R_1 + R_2} \right)}$$

$$\tau = \frac{L_1 L_2 (R_1 + R_2)}{(L_1 + L_2) (R_1 R_3 + R_2 R_3 + R_1 R_2)}$$